

**THE THIRTY-SEVENTH WILLIAM LOWELL PUTNAM
MATHEMATICAL COMPETITION**

December 4, 1976

A-1. P is an interior point of the angle whose sides are the rays OA and OB . Locate X on OA and Y on OB so that the line segment XY contains P and so that the product of distances $(PX)(PY)$ is a minimum.

A-2. Let $P(x, y) = x^2y + xy^2$ and $Q(x, y) = x^2 + xy + y^2$. For $n = 1, 2, 3, \dots$, let $F_n(x, y) = (x + y)^n - x^n - y^n$ and $G_n(x, y) = (x + y)^n + x^n + y^n$. One observes that $G_2 = 2Q$, $F_3 = 3P$, $G_4 = 2Q^2$, $F_5 = 5PQ$, $G_6 = 2Q^3 + 3P^2$. Prove that, in fact, for each n either F_n or G_n is expressible as a polynomial in P and Q with integer coefficients.

A-3. Find all integral solutions of the equation

$$|p^r - q^s| = 1,$$

where p and q are prime numbers and r and s are positive integers larger than unity. Prove that there are no other solutions.

A-4. Let r be a root of $P(x) = x^3 + ax^2 + bx - 1 = 0$ and $r + 1$ be a root of $y^3 + cy^2 + dy + 1 = 0$, where a, b, c , and d are integers. Also let $P(x)$ be irreducible over the rational numbers. Express another root s of $P(x) = 0$ as a function of r which does not explicitly involve a, b, c , or d .

A-5. In the (x, y) -plane, if R is the set of points inside and on a convex polygon, let $D(x, y)$ be the distance from (x, y) to the nearest point of R . (a) Show that there exist constants a, b , and c , independent of R , such that

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-D(x,y)} dx dy = a + bL + cA,$$

where L is the perimeter of R and A is the area of R . (b) Find the values of a, b , and c .

A-6. Suppose $f(x)$ is a twice continuously differentiable real valued function defined for all real numbers x and satisfying $|f(x)| \leq 1$ for all x and $(f(0))^2 + (f'(0))^2 = 4$. Prove that there exists a real number x_0 such that $f(x_0) + f''(x_0) = 0$.

B-1. Evaluate

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left(\left\lfloor \frac{2n}{k} \right\rfloor - 2 \left\lfloor \frac{n}{k} \right\rfloor \right)$$

and express your answer in the form $\log a - b$, with a and b positive integers.

Here $\{x\}$ is defined to be the integer such that $\{x\} \leq x < \{x\} + 1$ and $\log x$ is the logarithm of x to base e .

B-2. Suppose that G is a group generated by elements A and B , that is, every element of G can be written as a finite "word" $A^{n_1} B^{n_2} A^{n_3} \cdots B^{n_k}$, where $n_1, \dots, n_k$ are any integers, and $A^0 = B^0 = 1$ as usual. Also, suppose that $A^4 = B^7 = ABA^{-1}B = 1$, $A^2 \neq 1$, and $B \neq 1$.

(a) How many elements of G are of the form C^2 with C in G ?

(b) Write each such square as a word in A and B .

B-3. Suppose that we have n events $A_1, \dots, A_n$, each of which has probability at least $1 - a$ of occurring, where $a < 1/4$. Further suppose that A_i and A_j are mutually independent if $|i - j| > 1$, although A_i and A_{i+1} may be dependent. Assume as known that the recurrence $u_{k+1} = u_k - au_{k-1}$, $u_0 = 1$, $u_1 = 1 - a$, defines positive real numbers u_k for $k = 0, 1, \dots$. Show that the probability of all of $A_1, \dots, A_n$ occurring is at least u_n .

B-4. For a point P on an ellipse, let d be the distance from the center of the ellipse to the line tangent to the ellipse at P . Prove that $(PF_1)(PF_2)d^2$ is constant as P varies on the ellipse, where PF_1 and PF_2 are the distances from P to the foci F_1 and F_2 of the ellipse.

B-5. Evaluate

$$\sum_{k=0}^n (-1)^k \binom{n}{k} (x-k)^n.$$

B-6. As usual, let $\sigma(N)$ denote the sum of all the (positive integral) divisors of N . (Included among these divisors are 1 and N itself.) For example, if p is a prime, then $\sigma(p) = p + 1$. Motivated by the notion of a "perfect" number, a positive integer N is called "quasiperfect" if $\sigma(N) = 2N + 1$. Prove that every quasiperfect number is the square of an odd integer.